# Digital Material Passport

*ID* **DMP-2024-ARR-001** - *Version* **0.1.0**
*Issue Date* **2024-12-15** - *Certificate Type* **EN 10204 3.1**

| Manufacturer | Customer | Business Transaction |
|---|---|---|
| **Advanced Steel Technologies GmbH**<br>Industriestraße 15<br>52070 Aachen<br>DE<br>quality@advanced-steel.de | **Precision Engineering Ltd.**<br>Manufacturing Park 42<br>Birmingham<br>B1 1AA<br>GB<br>procurement@precision-eng.co.uk | *Order ID* **PO-2024-7890**<br>*Delivery ID* **DN-2024-3456** - Quantity **500 kg** |

## Product

| | | | |
|---|---|---|---|
| *Product Name* | Hardenability Test Steel | *Shape* | RoundBar - D 100 mm - Length 1000 mm |
| *Batch ID* | HTB-2024-045 | | |

## Chemical Analysis

*Heat Number* **H-2024-078** - *Sample Location* **Ladle**

| *Symbol* | *Unit* | *Min* | *Max* | *Actual* |
|---|---|---|---|---|
| **C** | % | 0.42 | 0.5 | 0.45 |
| **Mn** | % | 0.5 | 0.8 | 0.65 |
| **Si** | % | | 0.4 | 0.25 |
| **P** | % | | 0.03 | 0.015 |

| *Symbol* | *Unit* | *Min* | *Max* | *Actual* |
|---|---|---|---|---|
| **S** | % | | 0.025 | 0.008 |
| **Cr** | % | | 0.25 | 0.12 |
| **Ni** | % | | 0.25 | 0.08 |

## Mechanical Properties

| *Property* | *Symbol* | *Actual* | *Minimum* | *Maximum* | *Method* | *Status* |
|---|---|---|---|---|---|---|
| **Jominy Hardenability Test**<br>Quenched from 850°C | | | | | ISO 377 | ✓ |

| Distance from - quenched end (mm) | **1.5** | **3** | **5** | **7** | **9** | **11** | **15** | **20** | **25** | **30** |
|---|---|---|---|---|---|---|---|---|---|---|
| Value [HRC] | 45 | 43 | 40 | 36 | 33 | 31 | 27 | 25 | 28 | 27 |
| Min | 42 | 39 | 35 | 32 | 29 | 26 | 22 | 20 | | |
| Max | 47 | 46 | 44 | 41 | 39 | 37 | 33 | 31 | <= 30 | <= 29 |
| Status | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass |

| *Property* | *Symbol* | *Actual* | *Minimum* | *Maximum* | *Method* | *Status* |
|---|---|---|---|---|---|---|
| **Coating Thickness Profile**<br>Magnetic induction method | | | | | ISO 2178 | ✓ |

| Position along length (mm) | **0** | **50** | **100** |
|---|---|---|---|
| Value [µm] | 125 | 115 | 108 |
| Min | >= 120 | >= 110 | >= 100 |
| Max | <= 130 | <= 125 | <= 120 |
| Status | Pass | Pass | Pass |

## Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated by **Dr. Maria Schmidt**, Metallurgical Engineer, Quality Control Laboratory - 2024-12-15